

Program: Diploma in Microelectronics	
Course Code: 6499B	Course Title: Robotics Lab
Semester: 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To acquire the knowledge on advanced algebraic tools for the description of motion.
- To analyze and design the motion for articulated systems
- To develop an ability to use software tools for analysis and design of robotic systems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Experiments using Arduino		Introduction to IoT Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Demonstrate the mechanism of various robots	15	Applying
CO2	Interface basic robot with different sensors	15	Applying
CO3	Apply principles of kinematics to generate workspaces of planar and spatial robots	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3		3	2
CO2	3	3	3	3		3	2
CO3	3	3	3	3		3	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the mechanism of various robots		
M1.01	Simulation of Mathematical Model of Robot	3	Applying
M1.02	Simulate the robot motion for various inputs of the joint angular value for Move master RM-501 robotic arm	3	Applying
M1.03	Construct a robotic arm using servo motors or stepper motors and program the arm to perform various tasks, such as picking up objects, sorting the colour, or drawing shapes.	3	Applying
M1.04	Solve direct kinematics of a PUMA 560 robotic arm for the given joint variables and simulate the robot remotely to identify the end-effector trajectory	3	Applying
M1.05	Determine the joint variables of a PUMA 560 robotic arm for a given	3	Applying
CO2	Interface basic robot with different sensors		
M2.01	Simulating a Differential Drive Robot Using ROS	3	Applying
M2.02	Interfacing of DC geared motor with Robot Controller.	3	Applying
M2.03	Interfacing of quadrature encoder with Robot Controller	3	Applying

M2.04	Interfacing of ultrasonic distance sensors to the Robot Controller	3	Applying
M2.05	Interfacing of IR proximity sensor to the Robot Controller	3	Applying
M2.06	Write a simple program to control a robot's motion (e.g., moving forward, backward, turning) using programming commands.	3	Applying
	Lab Exam	1.5	
CO3	Apply principles of kinematics to generate workspaces of planar and spatial robots		
M3.01	Generate workspace of planar robots	3	Applying
M3.02	Generate workspace of spatial robots	3	Applying
M3.03	Forward dynamic analysis of a 2R planar robot	3	Applying
M3.04	Inverse dynamic analysis of a 2R planar robot	3	Applying
	Lab Exam	1.5	

****Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.)

Example:

1. Forward kinematic analysis of a 2R planar robot and 6 DoF robots
2. Inverse kinematic analysis of a 2R planar robot and 6 DoF robots

Text /Reference:

T/R	Book Title/Author
T1	Saha, S.K., "Introduction to Robotics, 2nd Edition, McGraw-Hill Higher Education, New Delhi, 2014.
T2	R.K.Mittal and I.J.Nagrath, Robotics and Control, Tata McGraw Hill, New Delhi, 4th Reprint, 2017.
R1	R.D.Klafter, T.A.Chimielewski and M.Negin, Robotic Engineering–An Integrated Approach, Prentice Hall of India, New Delhi, 1994.
R2	Mikell.P.Groover, Mitchell Weiss, Roger.N.Nagel, Nicholas.G.Odrey, "Industrial Robotics-Technology, Programming and Applications", Tata McGraw-Hill Publishing Company Limited, New Delhi, Third Reprint 2008

R3	B.K.Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied Publishers, Chennai, 1998.
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Online Resources:

Sl. No	Website Link
1	http://vlabs.iitkgp.ernet.in/mr/index.html# (IIT Kharagpur)
2	RoboAnalyzer (IIT Delhi)
3	RT Toolbox (Mitsubishi Robots)